



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

October 19, 2022

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 14833
(ELEVENTH REVISION)

EXPIRATION DATE: 2026-08-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Joyson Safety Systems Aschaffenburg GmbH
Aschaffenburg, Germany

U.S. AGENT: Joyson Safety Systems Acquisitions LLC
Armada, MI

2. PURPOSE AND LIMITATIONS:

a. This special permit authorizes the manufacture, mark, sale, and use of a non-DOT specification cylinder (pressure vessel) for use as a component for safety systems and explosive articles. The pressure vessel, charged with non-toxic, non-liquefied gases, is authorized for transportation in commerce subject to the requirements and limitations specified herein. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.

b. FIFTEEN-YEAR TRANSPORTATION AUTHORIZATION: This special permit authorizes transportation of the pressure vessel identified herein for up to fifteen years from the date of manufacture. This limitation does not apply to non-specification pressure vessels when installed in air bag modules. This special permit provides no certification of safety for end use environments and life cycles.

c. SPECIAL PERMIT SCOPE LIMITATIONS: This special permit only applies to a package of pressure vessels or seat-belt pretensioners when it is an article of commerce in transportation. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use, incorporation as a component of a vehicle or other device, other uses not associated with transportation in commerce.

- d. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing special permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.301(a) and 173.302(a)(2), and § 178.65(f)(2) in that a non-DOT specification cylinder is not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Joyson Safety Systems Aschaffenburg GmbH dated September 2, 2022, submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Material Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Non-toxic, non-flammable non-liquefied gases and mixtures thereof/Proper shipping name as specified in § 172.101	2.2	As appropriate	N/A
Safety devices, <i>electrically initiated</i>	9	UN3268	N/A
Safety devices, pyrotechnic	1.4G	UN0503	N/A

NOTE: A component which contains a quantity of pyrotechnic materials must be classed and approved as provided for in § 173.56 or § 173.166 of the HMR, as appropriate. If the pyrotechnic material augments the volume of the gas in the pressure vessel, or in any way enhances the performance of the compressed gas, the device must be tested in the same configuration as when shipped.

7. SAFETY CONTROL MEASURES:
- a. PACKAGING: Prescribed packaging is a non-DOT specification cylinder as shown in the Joyson Safety Systems drawings on file with the Office of Hazardous Materials Safety and meets the following requirements:
- (1) The maximum service pressure at 70 °F may not exceed 9,434 psig. The minimum test pressure is the pressure of the contents at 212 °F. The rated service pressure may not exceed 80% of the test pressure and the water volume of each pressure vessel may not exceed one liter. The term “pressure of contents” as used

in this special permit means the total pressure of all the materials to be shipped in the pressure.

(2) Material of construction must conform to all requirements of § 178.65(b).

(3) Manufacturing requirements must conform to all requirements of § 178.65(c), except as follows:

(i) Controlled marking of the sidewall by laser etching or other electro-mechanical means is permitted provided the depth of the marking does not exceed 0.22 mm (0.009”), and the height of the marking must be between 2.54 mm (0.10”) and 6.35 mm (0.26”).

(ii) Welding of attachments to the cylinder after the proof test performed in accordance with paragraph 7.b.(1) of this special permit is permitted. Following the welding of the attachment, each cylinder must have a leak test performed in accordance with American Society for Testing and Materials standard ASTM E493, method B.

(4) The minimum wall thickness must be such that the wall stress meets the requirements of § 178.65(d).

(5) Openings and attachments must conform to all requirements of § 178.65(e), except as follows:

(i) The diameter of the circle as referenced in § 178.65(e) may not exceed 100%.

(ii) Sidewall openings and attachments are authorized, provided the openings and attachments are not detrimental to the cylinder’s structural integrity. The burst performance of the cylinder shall not be less than 95% of the same cylinder design without attachment. In all cases, the burst pressure must meet the criteria of paragraph 7.b.(4). This must be demonstrated through certification by the Independent Inspection Agency as provided in paragraph 8.d.

(6) Each pressure vessel must be equipped with a pressure relief device designed to meet all the requirements for a rupture disk prescribed in the Compressed Gas Association (CGA) Pamphlet S-1.1. The pressure relief device must be capable of preventing rupture of the pressure vessel when subjected to a fire test conducted in accordance with CGA Pamphlet C-14 or tested in accordance with Test Series 6(c) of Part I of the UN Manual of Tests and Criteria.

b. TESTING:

- (1) Each pressure vessel must be tested as required in § 178.65(f), except that:
 - (i) Each cylinder must be tested at an internal pressure of at least the test pressure and must be held at that pressure for at least 30 seconds, or it may be limited to that which is adequate to ensure compliance with 178.65(f)(i) to (iii); and
 - (ii) The maximum duration of the shift specified in § 178.65(f)(3) may be extended beyond the 10 hours at the discretion of the independent inspector.
- (2) The flattening specified in § 178.65(g) is not required.
- (3) A representative pressure vessel, packaged as it would be for shipment, must be activated and no materials other than non-toxic, non-flammable vapors or gases may be expelled from the package.
- (4) A cylinder with an outside diameter less than two (2) inches may burst at a location other than specified in §§ 178.65(f)(2)(ii) and (iii). Additionally, all cylinders must have a burst pressure greater than or equal to 2.5 times the service pressure.

c. MARKING: Each pressure vessel must be durably marked as follows:DOT SP-14833/9434¹Serial No. xxxxx²

Manufacturer's Name

This³ Pressure Vessel May Not Be Refilled¹Where 9434 represents the service pressure.²Where xxxxx is the serial number as appropriate.³Where “This” and “The” may be used interchangeably.

Note 1: Each line of these markings may be placed without regard to location or order on the pressure vessel. The serial number must be traceable to the production lot under which the cylinder was manufactured in accordance with the requirements specified in §§ 178.65(f)(3) and 178.65(i).

Note 2: Standardization of the cylinder marking to use “This” must be completed by the next renewal date for this special permit.

8. SPECIAL PROVISIONS:

- a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.
- b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.
- c. This special permit is limited to pressure vessels used as components for safety systems and explosive articles. The pressure vessels are excepted from the requirements of the HMR, Part 178 when the design has been certified by an Independent Inspection Agency, approved under 49 CFR 107 as having met all of the requirements of this special permit.
- d. The Independent Inspection Agency's design certification must include test results and documents explosive classification and approval. A copy of the certification must be maintained at each facility where the pressure vessel is manufactured and by the Independent Inspection Agency for a period of 15 years from the date of completion of the design certification.
- e. A current copy of this special permit must be maintained at each facility where the pressure vessel is offered or reoffered for transportation.
- f. A current copy of this special permit must be maintained at each facility where the pressure vessel is manufactured under this special permit. It must be made available to a DOT representative upon request.
- g. Joyson Safety Systems Aschaffenburg GmbH must comply with all provisions of this special permit, and all other applicable requirements contained in the 49 CFR, Parts 100-180. No modification may be made to the pressure vessel or pyrotechnic components which would affect the performance of the pressure vessel or its compliance with the requirements of the special permit until such modifications have been reviewed, tested and certified by an Independent Inspector as meeting the requirements of this special permit.
- h. Devices utilizing the non-DOT specification pressure vessel authorized herein are exempt from the requirements of CFR 49 Parts 100-185 when installed in a safety components such as steering columns or door panels.
- i. Pressure vessels must be transported in strong outside packaging in accordance with § 173.301(a)(9), unless otherwise directed by §§ 173.166 or 173.62.

- j. Transportation of a Division 2.1 material (flammable gas) is not authorized aboard cargo vessel or aircraft unless specifically authorized in the Hazardous Material Table in (§ 172.101).
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, passenger-carrying aircraft (may not exceed the quantity limitation specified in § 172.101, Column (9A)), and cargo-only aircraft. (See restriction in paragraph 8.j. above).
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel, aircraft or motor vehicle used to transport packages covered by this special permit. The shipper must furnish a copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)—“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this

special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at

<https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: CWF/TG